Entry type: **FAQ** Entry ID: **109480387**, Entry date: **01/31/2017**

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How do you implement module and channel diagnostics in the user program of the SIMATIC S7-1200/S7-1500?

Entry | Associated product(s)

This entry provides a function block for diagnosing channel errors in the central configuration or in an IO module connected via PROFINET IO. This block uses the "RALRM" instruction.

You implement module and channel diagnostics of IO modules that support diagnostics (like the function classes ST and HF of the ET 200SP) by calling the "RALRM" instruction in the diagnostic interrupt OB 82.
Module and channel diagnostics via the SIMATIC S7-1200/S7-1500 is illustrated taking the example of the central IO and via PROFINET IO:

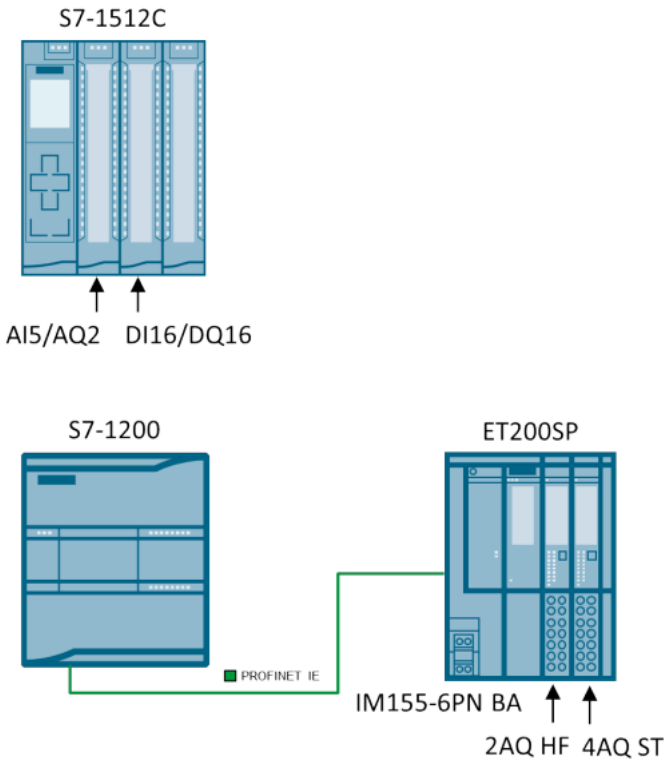

Fig. 1

Using the FB "1x00Diag82" and depending on the controller family used (S7-1200 or S7-1500) you can decode and process per module or channel the diagnostics data of the "RALRM" instruction for the central or distributed configuration via PN IO.

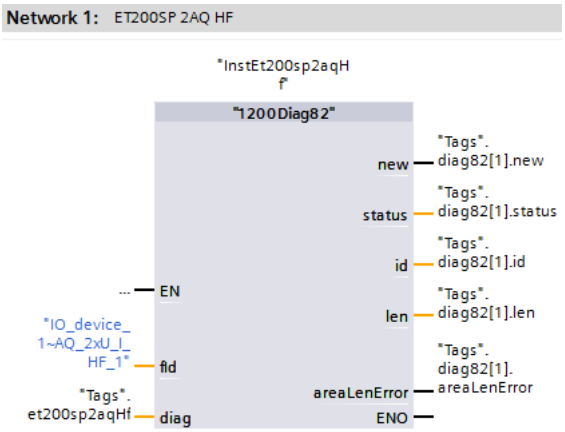

Fig. 2

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Documentation (941,9 KB) (/cs/attachments/109480387/109480387_Diag82_v20_en.pdf)
TIA Portal Project V14 Upd1 (657,7 KB) (/cs/attachments/109480387/109480387_Diag82_v20.zip)
TIA Portal Project V15 (1,2 MB) (/cs/attachments/109480387/109480387_Diag82_v20_V15.zip) for TIA Portal V15 or higher.

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Further Information

- FAQ "What are the differences between the Basic (BA), Standard (ST), High Feature (HF) and High Speed (HS) modules of the ET 200SP and ET 200MP" (/cs/document/109476914/what-are-the-differences-between-the-basic-(ba)-standard-(st)-high-feature-(hf)-and-high-speed-(hs)-modules-of-the-et-200sp-and-et-200mp-?lc=en-ww)
- Application example "Diagnostics in the User Program with S7-1500" (/cs/document/98210758/diagnostics-in-user-program-with-s7-1500?lc=en-ww)
- Application example "Diagnostics in the User Program with S7-1200" (/cs/document/109483251/diagnostics-in-user-program-with-s7-1200?lc=en-ww)

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- Automation Technology > Automation Systems > Industrial Automation Systems SIMATIC > PLC > Basic Controller > S7-1200 > Central processing units > Fail-safe CPUs (products?pnid=13686)
- Automation Technology > Automation Systems > Industrial Automation Systems SIMATIC > PLC > Advanced Controller > S7-1500 > CPUs > Standard-CPUs (products?pnid=13718)
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- Automation Technology > Industry software > Automation software > TIA Portal > PLC programming > STEP 7 Basic (TIA Portal) (products?pnid=14674)
- Automation Technology > Products for specific requirements > Automatision technology for extreme conditions > SIPLUS extreme Industrial Automation Systems > Controllers > SIPLUS Basic Controller > SIPLUS S7-1200 > SIPLUS Central Processing Units > SIPLUS Standard CPUs (products?pnid=18082)
- Automation Technology > Products for specific requirements > Automatision technology for extreme conditions > SIPLUS extreme Industrial Automation Systems > Controllers > SIPLUS Advanced Controller > SIPLUS S7-1500 > SIPLUS Central Processing Units > SIPLUS Standard CPUs (products?pnid=18121)
- Automation Technology > Automation Systems > Industrial Automation Systems SIMATIC > PLC > Advanced Controller > S7-1500 > CPUs > Compact CPUs (products?pnid=21360)
- Automation Technology > Automation Systems > Industrial Automation Systems SIMATIC > Software for SIMATIC Controller > STEP 7 (TIA Portal) (products?pnid=24471)

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